

FR-12: A matching-indexed doubling counterexample

George Stepaniants

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, California, USA

12 September 2026

Abstract

Let $H(n)$ count labeled real Hadamard matrices of order n . An injective construction gives $H(2m) \geq (2m-1)!!H(m)^2$. Iteration yields $H(2^k) \geq 2^{2^k(k-1)(k-2)/8}$ for $k \geq 2$. Consequently no absolute constant C satisfies $H(n) \leq 2^{Cn \log_2 n}$ at every order divisible by four. This refutes the exact counting target of FR-12; it does not address existence at every admissible order.

Assistance and independent review. Prepared with substantial assistance from Codex. A separate Codex agent independently audited the complete supplied argument and reported PASS on 12 September 2026. The signed independent review identifies the exact archived source and scope. This is an informal AI-agent audit, not external human peer review or formal verification. The supplementary finite checker is not a universal proof certificate, and no historical priority certification is claimed.

1 Exact target

Write

$$\mathcal{H}_n = \{A \in \{-1, 1\}^{n \times n} : AA^\top = nI_n\}, \quad H(n) = |\mathcal{H}_n|.$$

Rows and columns are labeled; signed permutations are *not* quotiented out. The target in FR-12 [2], attributed to Ferber–Jain–Zhao [1], is whether an absolute $C > 0$ gives

$$H(n) \leq 2^{Cn \log_2 n} \quad (4 \mid n). \quad (1)$$

We count actual matrices, so no assumption about automorphism groups or sizes of equivalence classes enters the argument.

Lemma 1 (Injective doubling). *For every positive integer m ,*

$$H(2m) \geq \frac{(2m)!}{2^m m!} H(m)^2 = (2m-1)!! H(m)^2. \quad (2)$$

Proof. Let $\mathfrak{M}_{2m}$ denote the perfect matchings of the labeled positions $\{1, \dots, 2m\}$. Given a matching, write each pair as (u_i, v_i) with $u_i < v_i$, and order its pairs by $u_1 < \dots < u_m$. For $A, B \in \mathcal{H}_m$, define a $2m \times 2m$ sign matrix C by

$$C_{u_i, \cdot} = (A_{i, \cdot}, B_{i, \cdot}), \quad C_{v_i, \cdot} = (A_{i, \cdot}, -B_{i, \cdot}), \quad 1 \leq i \leq m. \quad (3)$$

Each row has squared norm $2m$. The inner product of the two rows in one pair is $m - m = 0$. For two different pairs $i \neq j$, all four possible inner products vanish, since they have the form

$$\langle A_{i,\cdot}, A_{j,\cdot} \rangle \pm \langle B_{i,\cdot}, B_{j,\cdot} \rangle = 0.$$

Thus $C \in \mathcal{H}_{2m}$.

The map $(A, B, \mathcal{M}) \mapsto C$ is injective. Indeed, distinct rows of A are distinct vectors: equality of two rows would contradict their orthogonality. It follows that, among the rows of C , equality of the *first m coordinates* partitions the $2m$ row positions into exactly the pairs of $\mathcal{M}$. Hence C recovers the matching. Its canonical pair ordering then recovers A from these first halves. For each pair, the last m coordinates of the smaller-indexed row recover the corresponding row of B . This also fixes every sign, so no residual multiplicity remains.

Finally, $|\mathfrak{M}_{2m}| = (2m)!/(2^m m!)$: list all positions in an order, pair successive positions, and divide by the 2^m orders within pairs and the $m!$ orders of the pairs. Counting the domain of the injection proves (2). $\square$

Theorem 1 (A super- $n \log n$ exponent). *For every integer $k \geq 2$,*

$$\boxed{H(2^k) \geq 2^{2^k(k-1)(k-2)/8}}. \quad (4)$$

In particular, the assertion (1) is false.

Proof. Hadamard matrices exist at every power of two, by applying (3) recursively starting with an order-one sign matrix. Consequently the logarithms below are defined. Put

$$a_k = 2^{-k} \log_2 H(2^k).$$

For $k \geq 2$, let $m = 2^{k-1}$, so m is even. The matching factor satisfies

$$(2m-1)!! = \prod_{j=1}^m (2j-1) \geq m! \geq (m/2)^{m/2}.$$

The last inequality retains just the largest $m/2$ factors in $m!$ and bounds each from below by $m/2$. Taking logarithms of (2) and dividing by $2m$ gives

$$a_k \geq a_{k-1} + \frac{\log_2 m - 1}{4} = a_{k-1} + \frac{k-2}{4}.$$

Since $a_1 \geq 0$, summation yields

$$a_k \geq \frac{1}{4} \sum_{j=2}^k (j-2) = \frac{(k-1)(k-2)}{8},$$

which proves (4). If (1) held, applying it at $n = 2^k$ would give $a_k \leq Ck$. This contradicts the displayed quadratic lower bound for sufficiently large k . All these n are divisible by four. $\square$

2 Checks, attribution, and scope

The supplementary script checks the equivalent row-permutation formulation: form $\begin{pmatrix} A & B \\ A & -B \end{pmatrix}$, then permute its $2m$ rows. Every output has exactly $2^m m!$ preimages in that formulation. The matching-indexed construction in Lemma 1 removes this multiplicity explicitly.

m	$H(m)$	Input triples	Distinct outputs	Fiber size
1	2	8	4	2
2	8	1536	192	8

Those finite cases, and exact integer checks of the iterated lower bound through $k = 10$, are not used to infer an asymptotic statement. The analytic injection and its iteration establish the result for every k .

The construction uses only power-of-two orders. It neither resolves the Hadamard existence conjecture nor supplies a matching upper bound on $H(n)$. The original counting question and the published upper bounds remain attributed to their sources. The accompanying audit states the observed public branch scope and the limits of the duplicate/literature search.

References

- [1] A. Ferber, V. Jain, and Y. Zhao, *On the number of Hadamard matrices via anti-concentration*, *Combinatorics, Probability and Computing* 31 (2022), 455–477. doi:10.1017/S0963548321000377. Published Conjecture 1.3; Conjecture 1.4 in arXiv:1808.07222v1.
- [2] *Open Problems in Numerical Linear Algebra*, entry FR-12, `ajt60gaibb/OpenProblemsInNLA`, snapshot `1f22006bdaa4659fcaa0bb775a887685cd3cc566`.
<https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/1f22006bdaa4659fcaa0bb775a887685cd3cc566/frames-and-matrix-designs/FR-12/README.md>.